A DECODE ROUND SUBMISSION

On

ADAPTIVE LEARNING FOR CLASSES 6 TO 10

A teacher-led, concept-level adaptive learning platform aligned to NCERT and State Board syllabi

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1. PROBLEM STATEMENT

1.1 Chosen Problem

The chosen problem is **Adaptive Learning for Classes 6 to 10**. The task is to design a solution that understands where a student stands and changes what the student learns next, how it is learned, or how fast the student progresses, across all three parts of studying: learning a concept, solving problems and testing what is known. The solution must be built for Indian students and Indian classrooms, and must empower the school teacher instead of replacing the teacher.

1.2 Background

In a typical Indian classroom, a single teacher handles a large class that moves through the syllabus at one pace. Students who have already understood a topic wait for the others, while students who missed an earlier idea try to build new learning on a gap. In subjects such as Science and Mathematics, concepts are cumulative. For example, the Class 9 chapter on Motion depends on speed, units and graphs introduced in Classes 6 and 7, and Linear Equations in Class 9 depend on Algebraic Expressions in Class 8. A basic idea missed in Class 6 or 7 therefore makes every dependent topic harder later, and the effect is most severe in Classes 9 and 10, where board examinations leave little time to repair old gaps. Students also learn in languages that differ from the language of the textbook, attend tuition or coaching of varying quality, and have uneven access to devices and internet.

1.3 Root Cause

Learning gaps remain invisible until it is too late, and nothing connects a failed Class 9 question to the Class 6 or Class 7 idea that caused it. The contributing causes are:

- Syllabus, homework and tests are organised by chapter and grade, whereas learning actually depends on a network of prerequisite concepts that spans several grades.
- Assessment is mostly summative and arrives at the end of a chapter, so a gap is detected only after the class has moved on.
- A mark states how much a student scored, not why. A wrong answer may come from a missing prerequisite, a language barrier, a careless slip or a misconception, and each needs a different response.
- Teachers have no practical tool to see a concept-level picture of every student in a class of forty or more, and no time to prepare differentiated material.

1.4 Existing Systems and Disadvantages

1.4.1 Existing Systems

Students currently depend on textbooks and classroom teaching, chapter-end unit tests, private tuition and coaching, and digital platforms. Global adaptive learning platforms offer practice with mastery tracking, and recent AI tutors answer questions on demand. Most of these platforms are designed for other markets and report mainly to the student.

1.4.2 Disadvantages of Existing Systems

- Global platforms are not mapped to NCERT and State Board chapters, and they assume a personal device and reliable internet.
- Diagnosis stops at the wrong answer and does not trace the root gap through earlier grades.
- Difficulty is adjusted for a whole subject or grade, not for the single concept where the student is weak.
- The teacher is largely absent from the loop, so insight does not reach the person who can act on it.
- Coaching centres and AI tutors add study hours but not always understanding, and they draw students away from the school teacher.
- Support for regional languages, shared family phones and paper-based classrooms is limited.

Table 1.1 Stakeholders affected by the problem

Stakeholder	Present difficulty
Students (Classes 6 to 10, ages 11 to 16)	Build on missing basics and lose confidence. Fast learners are held back and become bored.
School teachers	Large classes, one syllabus pace, no concept-level view of each student and little time for individual attention.
Parents	See only marks and cannot tell what the child is struggling with. Many are more comfortable in a regional language.
Schools	Mixed languages, uneven devices and internet, and pressure to complete the syllabus on schedule.

2. PROPOSED SYSTEM

2.1 Overview

The proposed system is a school-based adaptive learning platform for Classes 6 to 10. The NCERT syllabus, and later State Board syllabi, are stored as a **concept graph** in which every concept is linked to its prerequisites and carries graded questions from basic to advanced. Each student has a working level for every concept. Weekly tests, short practice sets and teacher observations update these levels. When a student struggles, the system locates the root gap and the teacher decides on the remedy. The platform measures and recommends. The teacher teaches and decides.

2.2 Core Idea: Concept-Level Difficulty Adjustment

A grade-9 student is expected to handle the advanced questions of a chapter. If the student answers only the basic questions, the system does not simply report a low score. It walks backward through the prerequisite concepts until it finds the level at which the student answers consistently, and then **lowers the difficulty for that concept alone** to that level. All other chapters stay at the student's grade. As the teacher addresses the gap and the weekly tests show consistent success, the level rises one step at a time until the student reaches the grade target. The same ladder works upward for fast learners, who receive enrichment questions instead of repetition.

2.3 Worked Example

Student B is in Class 9 and takes the weekly test on Motion. B answers definition-level questions correctly but fails problems on uniform motion and word problems. The diagnostic probe presents a few questions from Class 8 and finds mixed results, then presents Class 7 questions on speed and graphs and again finds errors, and finally presents Class 6 questions, which B answers correctly. The root gap is therefore placed at Class 6 and 7 level, and B's working level for Motion is set accordingly (Figure 2.1). The teacher receives a report with the evidence, confirms it and assigns B to a remedial group. In the following weeks B takes the weekly tests at each step and moves up only after consistent results.

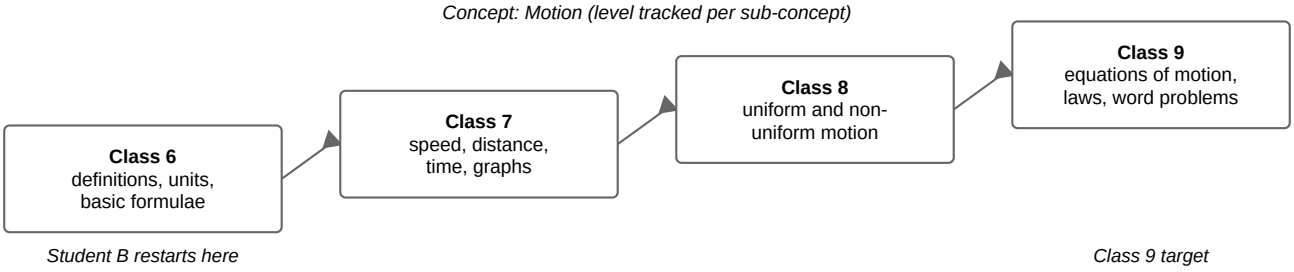

Figure 2.1: Concept ladder for Motion and the path of Student B

2.4 Adaptation Workflow

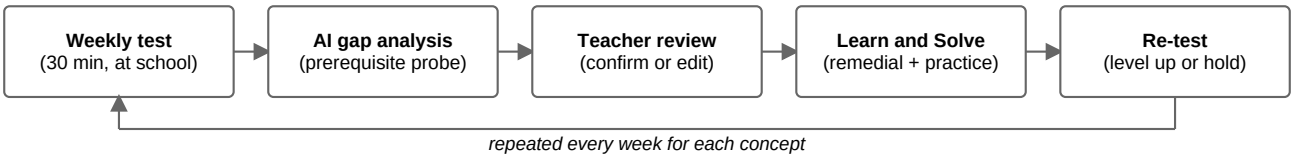

Figure 2.2: Weekly adaptation workflow of the system

2.5 Level Change Rules

- A level changes only after **two consistent results**, so a single bad day does not demote or block a student.

- Wrong answers are tagged by error type (prerequisite gap, calculation slip, misreading, misconception, language difficulty). Only prerequisite gaps and misconceptions trigger a level drop.
- If a student stalls twice at the same level, the teacher is alerted with a suggestion to change the explanation style instead of repeating the same lesson.
- Each weekly test contains about 60% questions at the working level, 25% prerequisite probes and 15% spaced review or stretch questions.
- The teacher can confirm, override or postpone any level change, and every change is logged.

Table 2.1 Adaptation across Learn, Solve and Test

Part	How the system adapts
Learn	Short micro-lessons of 5 to 10 minutes on the exact missing concept, in the student's language with English technical terms alongside. Remedial class outlines are generated for the teacher.
Solve	Practice sets rise or fall with the student's streaks and error type, so the next question matches the concept the student is currently building.
Test	A weekly 30-minute school test is generated per student from a tagged question bank. Different students receive different papers, which also reduces copying.

3. TEACHER EMPOWERMENT

3.1 Design Principle

The principle of the system is that **the AI advises and the teacher decides**. No student is promoted, demoted or grouped without the teacher seeing the supporting evidence. The platform takes over the measuring work (diagnosing, grading and tracking), which is where teachers lack time, and leaves explaining, motivating and judging to the teacher.

3.2 Teacher Functions

Table 3.1 Functions provided to the teacher

Teacher need	Function provided
See each student's real gap	AI gap report listing the missed questions, the error types and the likely root concept from an earlier grade.
Act in a large class	Automatic grouping of students with the same gap, for example 9 students needing speed and units, so one 30-minute session helps many. Editable lesson outline and printable worksheets at three levels.
Plan the day	Class heatmap by concept, a list of students needing a check-in, and students ready for a challenge.
Control the content	Teachers can add, edit and approve questions, and record oral or practical results that feed into the student profile.
Intervene early	Alerts for students who are stalling or declining, with a one-tap message to the parent.

3.3 Why Teachers Are Not Replaced

- Remedial teaching is delivered by the teacher in class. Micro-lessons only support that teaching.
- Weekly tests are taken at school under the teacher's supervision, which keeps results reliable and keeps the school at the centre of the student's learning.
- All AI-generated questions, explanations and summaries are reviewed by a teacher before students see them.

3.4 Benefits to Other Users

- **Student:** learns at the right level for each concept. Progress is shown as a path (for example, Foundation Path: Motion) and not as a lower grade label, and it is private from classmates.
- **Parent:** a separate login using a school-issued PIN with an OTP to the parent's phone, showing plain-language progress in the home language and suggestions for the week.
- **School:** concept-level view across sections, so remedial time is spent where gaps actually exist.

4. CONSTRAINTS

The system is designed around the conditions of Indian schools. Table 4.1 lists the main constraints and the response to each.

Table 4.1 Constraints and design responses

Constraint	Design response
Limited devices and data	Offline-first progressive web app with cached content and background sync. Shared classroom devices and multiple student profiles on a family phone. Paper-test fallback, where the teacher photographs or scans answer sheets.
Multiple boards and languages	Syllabus stored as a separate concept graph for each board. Content and hints in regional languages with English terms shown alongside, and voice read-out for slow readers.
Large classes, little time	Auto-grouping, printable worksheets and a one-screen daily plan, so insight costs minutes.
Privacy of minors	Parental consent, minimal data collection and school-controlled access, in line with the Digital Personal Data Protection Act, 2023.
Content volume	AI drafts questions and explanations, and teachers approve them. Start with one subject and expand chapter by chapter.
Stigma of low levels	Levels are never shown to classmates. Progress is presented as growth along a path.

5. SYSTEM APPROACH

5.1 Architecture

The system follows a three-tier architecture. The Presentation Layer contains the student app, teacher dashboard and parent view. The Application Layer holds the concept graph service, diagnostic and mastery engine, test generator and teacher insight service. The Database Layer stores the concept graph, question bank, student profiles, results and the offline sync queue (Figure 5.1).

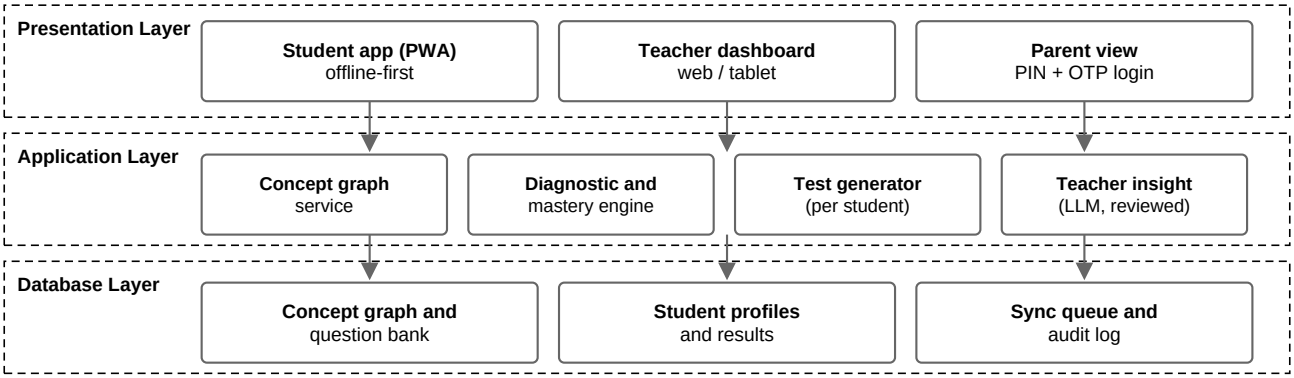

Figure 5.1: System architecture of the proposed system

5.2 System Modules

- **Concept Graph Module:** stores concepts, prerequisite links, grade mapping and board mapping.
- **Question Bank Module:** holds questions tagged by concept, level, language and error type, with a teacher approval status.
- **Placement and Diagnostic Module:** runs the initial adaptive placement test and the backward prerequisite probe.
- **Mastery Tracking Module:** maintains a per-student, per-concept mastery estimate using Bayesian Knowledge Tracing [1] and applies the level change rules.
- **Test Generation Module:** builds each weekly test from the student's working levels, probes and review items.
- **Teacher Dashboard Module:** shows gap reports, remedial groups, class heatmap, worksheets and alerts.
- **Parent Module:** secure progress view with suggestions in the home language.
- **Offline Sync and Paper-Test Module:** caches content, queues results and accepts scanned answer sheets.

Table 5.1 Proposed technology stack

Layer	Technology
Frontend	React progressive web app with service-worker caching, responsive layout, speech-to-text and text-to-speech for regional languages
Backend	Python (Flask or FastAPI) REST services
Database	PostgreSQL for profiles and results, with a graph structure for the concept map; IndexedDB on the device for offline data
AI components	Bayesian Knowledge Tracing for mastery, rule-based diagnostic probe, LLM for gap summaries and hint drafts (teacher-reviewed), OCR for scanned answer sheets

6. DEMONSTRATION FOCUS

The demonstration covers a narrow and deep slice, because a working slice shows adaptation, teacher empowerment and fit to India more convincingly than a broad claim.

Scope: Science for Classes 8 and 9, limited to one connected cluster (Motion, Force and Light), with about 100 to 150 questions tagged by concept, level and error type. **Scenario shown:** (1) placement test for Student B, (2) weekly test on Motion, (3) diagnosed root gap, (4) teacher reviews and approves, (5) remedial group formed and worksheet generated, (6) Motion level lowered to the correct step, (7) next weekly test and level-up. The teacher dashboard and the parent view for the same student are shown alongside.

7. SCALABILITY ACROSS INDIA

Three things change from board to board: the syllabus, the questions and the language. The platform keeps these in separate layers, so adding a curriculum means adding data and not rebuilding the product.

Table 7.1 Layers of curriculum scaling

Layer	Content	How it scales
Syllabus graph	Concepts, prerequisites and grade mapping per board	NCERT first, then Telangana and Andhra Pradesh SCERT and other State Boards. Concepts common to boards are shared, so only differences are mapped.
Question bank	Tagged questions by concept, level, language and error type	AI-drafted and teacher-approved. Approved questions can be shared between schools.
Language layer	Explanations, hints and voice	English, Telugu and Hindi first, then other languages without changing the learning logic.

The rollout proceeds in four phases: **Build** (working prototype for the demonstration cluster), **Pilot** (one or two schools, Science and Mathematics for Classes 8 and 9), **Expand** (Classes 6 to 10, a second board and more languages) and **Scale** (district and state partnerships and a teacher community for shared content). Success will be measured by the share of students who level up within a term, teacher time saved per week and weekly test completion rate. Principal risks are wrong diagnosis (handled by the two-result rule and teacher override), content quality (teacher review) and adoption (minimal teacher effort, printable and WhatsApp-shareable outputs, paper fallback).

8. CONCLUSION

The proposed system turns scattered chapter-wise marks into a concept-level picture of each student and places that picture in the hands of the teacher. By lowering difficulty only for the concept where a gap exists and raising it again as understanding is confirmed, it lets a Class 9 student repair a Class 6 gap without being held back in other subjects. Because it is designed around NCERT and State Board syllabi, regional languages, shared devices and paper-based classrooms, it can be adopted by ordinary Indian schools and extended to new boards by adding data.

9. REFERENCES

[1] A. T. Corbett and J. R. Anderson, “Knowledge tracing: Modeling the acquisition of procedural knowledge,” *User Modeling and User-Adapted Interaction*, vol. 4, no. 4, pp. 253–278, 1994.

[2] B. S. Bloom, “Learning for mastery,” *Evaluation Comment*, vol. 1, no. 2, pp. 1–12, 1968.

[3] Ministry of Education, Government of India, *National Education Policy 2020*, 2020.

[4] Government of India, *The Digital Personal Data Protection Act, 2023*, 2023.

[5] NCERT, *Science and Mathematics textbooks for Classes 6 to 10*, New Delhi.

[6] X-Kernel, GRIET, *Code Kernel '26 Problem Statement: Adaptive Learning for Classes 6 to 10*, Decode Round, 2026.